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International Mathematics Assessments for Schools

2022 / 2023 MIDDLE PRIMARY DIVISION FIRST ROUND PAPER

Time allowed：75 minutes

When your teacher gives the signal, begin working on the problems.

INSTRUCTION AND INFORMATION

GENERAL

1. Do not open the booklet until told to do so by your teacher.
2. No calculators, slide rules, log tables, math stencils, mobile phones or other calculating aids are permitted. Scribbling paper, graph paper, ruler and compasses are permitted, but are not essential.
3. Diagrams are NOT drawn to scale. They are intended only as aids.
4. There are 20 multiple-choice questions, each with 5 choices. Choose the most reasonable answer. The last 5 questions require whole number answers between 000 and 999 inclusive. The questions generally get harder as you work through the paper. There is no penalty for an incorrect response.
5. This is a mathematics assessment, not a test; do not expect to answer all questions.
6. Read the instructions on the answer sheet carefully. Ensure your name, school name and school year are filled in. It is your responsibility that the Answer Sheet is correctly coded.

THE ANSWER SHEET

1. Use only pencils.
2. Record your answers on the reverse side of the Answer Sheet (not on the question paper) by FULLY filling in the circles which correspond to your choices.
3. Your Answer Sheet will be read by a machine. The machine will see all markings even if they are in the wrong places. So please be careful not to doodle or write anything extra on the Answer Sheet. If you want to change an answer or remove any marks, use a plastic eraser and be sure to remove all marks and smudges.

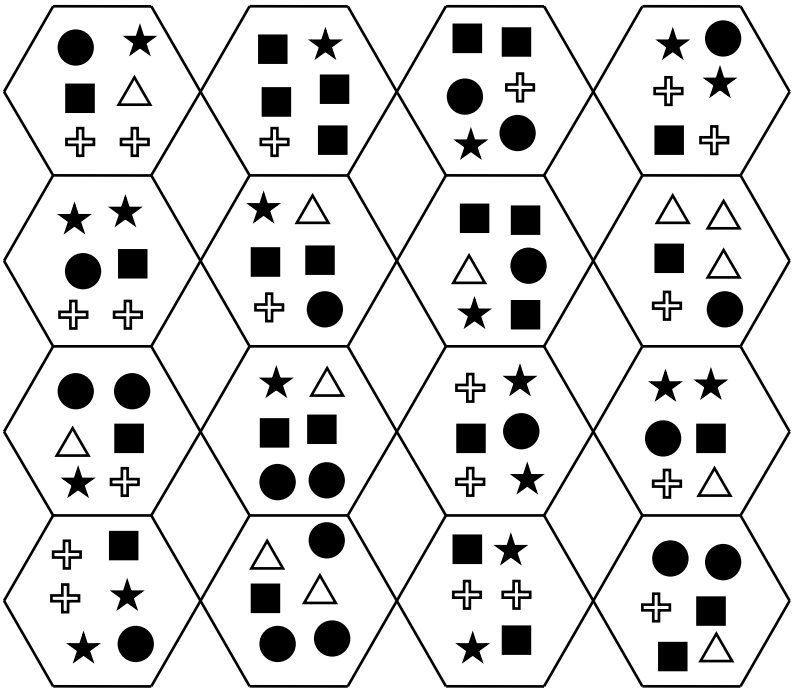
INTEGRITY OF THE COMPETITION

The IMAS reserves the right to re-examine students before deciding whether to grant official status to their scores.

2022 / 2023 MIDDLE PRIMARY DIVISION FIRST ROUND PAPER

Questions 1-10, 3 marks each

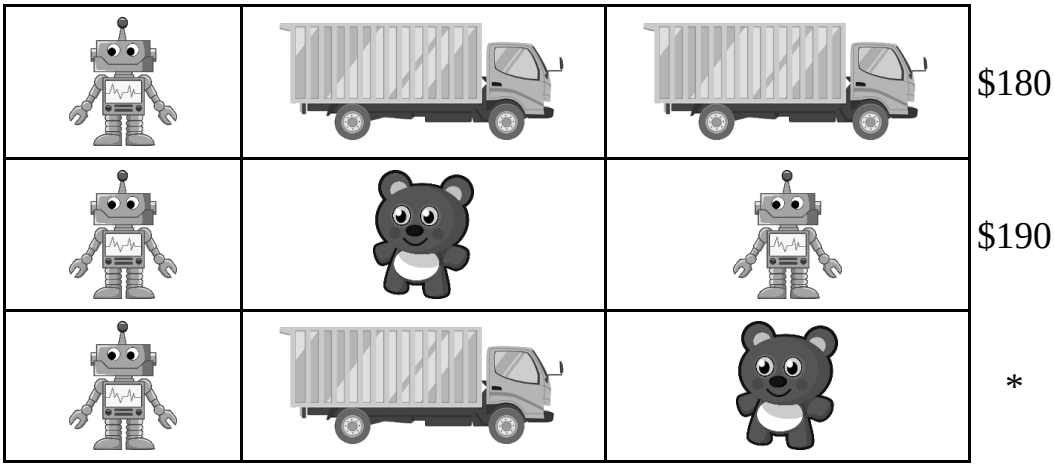
1. Among the 16 hexagons in the diagram below, what is the maximum number of hexagons that contain the same set of symbols, in the same number, not necessarily arranged in the same order?



（A）2　　（B）3　　（C）4　　（D）5　　（E）6

---

2. There are three kinds of toys and each one has a fixed price. The sums of the prices for each row and column are given, except for the last row and column, as shown in the diagram below.



What is the value of the symbol *?

（A）$160　　（B）$170　　（C）$200　　（D）$210　　（E）$220

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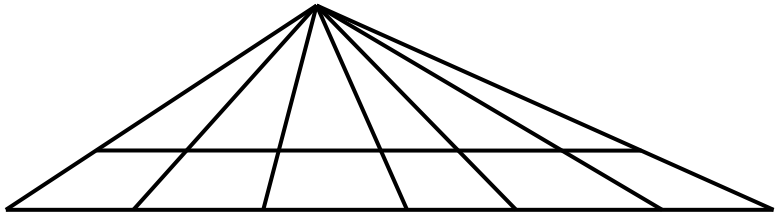
3. What is the value of the expression $(1234+2341+3412+4123)\div 5$?

（A）1111　（B）1234　（C）2022　（D）2222　（E）2468

4. Compute all possible products of two one-digit positive integers to construct a 9 by 9 multiplication table. How many two-digit numbers can be obtained in the table?

（A）14　（B）23　（C）58　（D）67　（E）81

5. How many different triangles can you find in the diagram below?

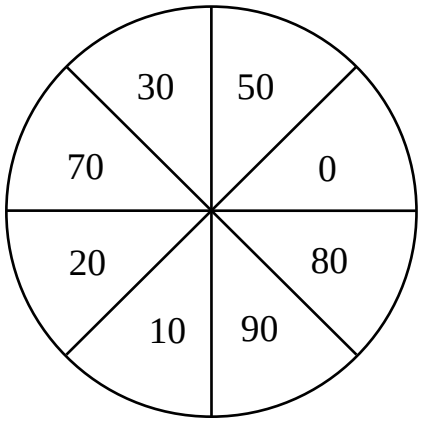


（A）50　（B）42　（C）30　（D）28　（E）24

6. Eight friends dined at a restaurant and decided to equally divide the amount of money that had to be paid. Because Tom left his money at home, his other seven friends paid an extra $25 each to cover his share. How much was the total bill of the eight friends?

（A）$175　（B）$200　（C）$800　（D）$1225　（E）$1400

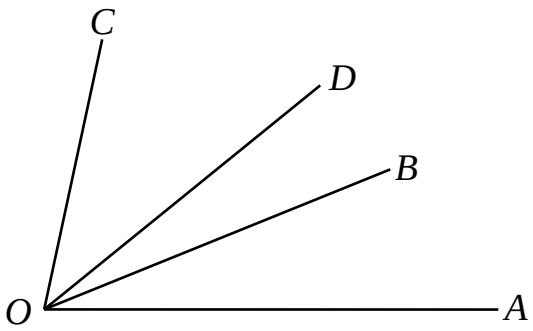
7. The diagram below shows a strange dartboard. If the dart hits one zone, the player scores the number of points written in that zone. If the dart hits the central point, it scores 90; if the dart hits the segment between two zones, it scores the higher score of the two zones.



Using three darts at a time, how many different ways are there of obtaining a total score of exactly 100? Note that all darts must be on the dartboard.

（A）4　（B）5　（C）6　（D）7　（E）8

8. There are three segments OA, OB and OC such that $\angle AOB = 22°$ and $\angle BOC = 56°$, as shown in the diagram below.



If $\angle AOC$ is split in two equal parts by the segment OD, what is the measure, in degrees, of $\angle BOD$?

（A）17　（B）18　（C）19　（D）20　（E）21

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9. Suppose that your height this year is 10% more than it was last year, and last year your height was 20% more than it was the year before. During the past two years, your height has increased by what percentage?

（A）30%　（B）31%　（C）32%　（D）34%　（E）40%

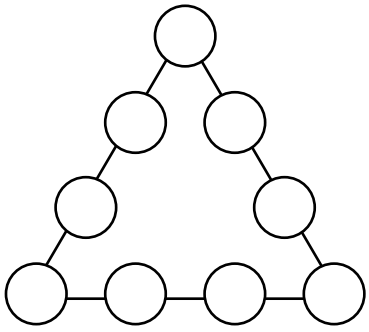
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10. The number 638*977 is a multiple of 7. How many different possible values of the middle digit represented by the symbol * are there?

（A）1　（B）2　（C）3　（D）4　（E）5

Questions 11-20, 4 marks each

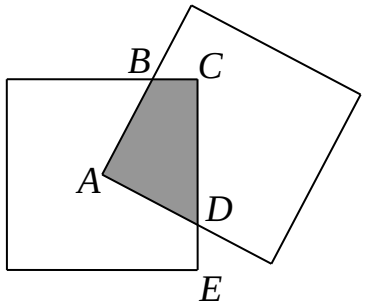
11. The digits 1, 2, 3, 4, 5, 6, 7, 8 and 9 are to be placed in the nine circles in the diagram exactly once such that the sums along each of the three lines are all equal. If each circle can only be filled by exactly one digit, what is the smallest value possible for this sum?



（A）15　（B）16　（C）17　（D）18　（E）19

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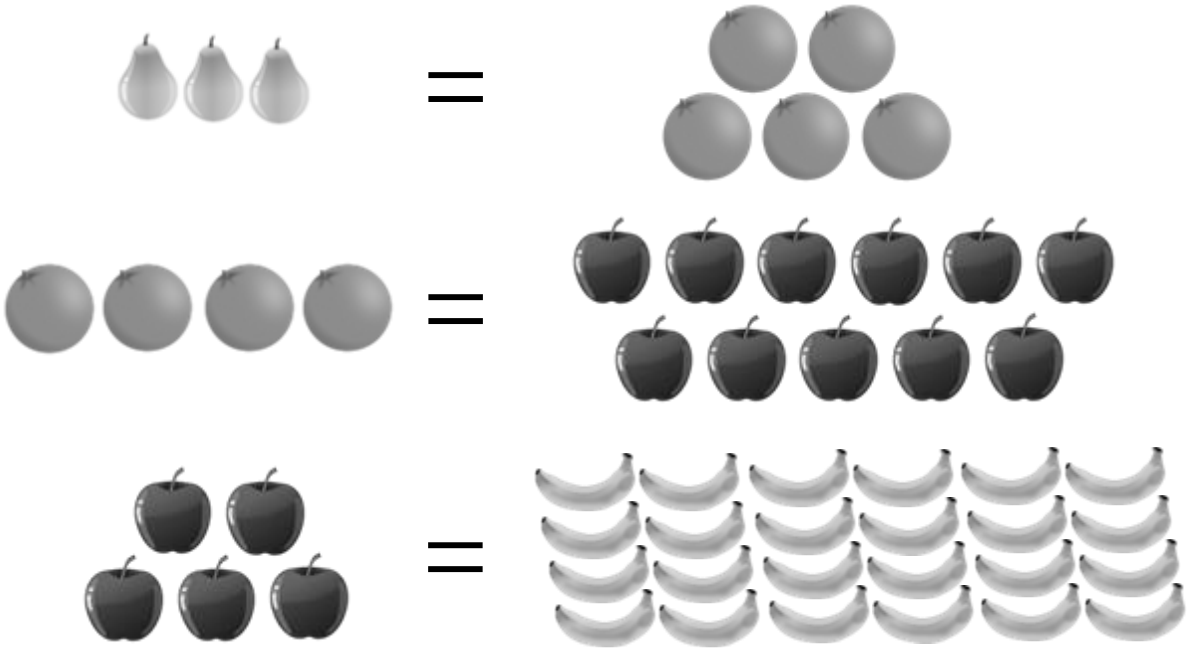
12. Two squares of side 8 cm intersect such that the corner of one square coincides with the centre of the other square, as shown in the diagram below.



If it is known that $BC = DE$, what is the area, in cm^2, of the shaded region?

（A）2　　（B）8　　（C）12　　（D）16　　（E）32

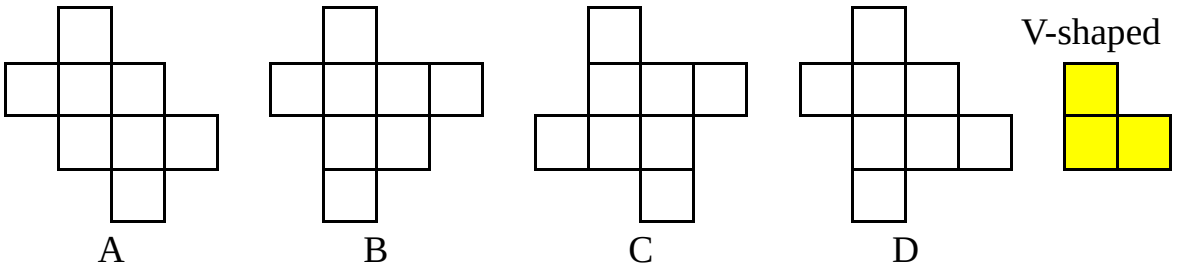
13. At a certain store, the cost of 3 avocados is the same as that of 5 oranges, the cost of 4 oranges is the same as that of 11 apples, and the cost of 5 apples is the same as that of 24 bananas.



What is the number of bananas that can be bought with the money for buying two avocados?

（A）22　　（B）44　　（C）264　　（D）42　　（E）48

14. The diagram below shows four diagrams, each consisting of eight unit squares, and a V-shaped piece consisting of three unit squares.



We use three copies of the V-shaped piece to try to cover each of the diagrams. The pieces may be turned or flipped, and they are allowed to overlap.
Which diagram **CANNOT** be covered?

（A）A only　（B）B only　（C）C only
（D）D only　（E）All of them can be covered

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15. How many odd multiples of 3 are there between 1800 and 2022?

（A）36　（B）37　（C）38　（D）74　（E）75

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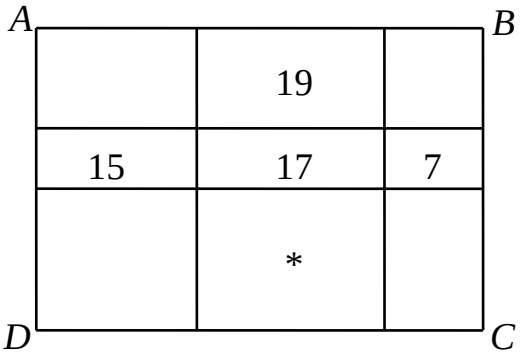
16. The average of 50 numbers is 38. If two of these numbers, namely 37 and 63, are discarded, what will be the average of the remaining numbers?

（A）36　（B）36.5　（C）37　（D）37.5　（E）38

---

17. A rectangle *ABCD* is divided into nine smaller rectangles as shown in the diagram below. The number written inside a small rectangle denotes its perimeter, in cm. Notice that the perimeter of one of the small rectangles has an unknown value, *. If the perimeter of rectangle *ABCD* is 50 cm, what is the value of the symbol *? Note that the diagram is not drawn to scale.

（A）20　（B）21　（C）23　（D）26　（E）29



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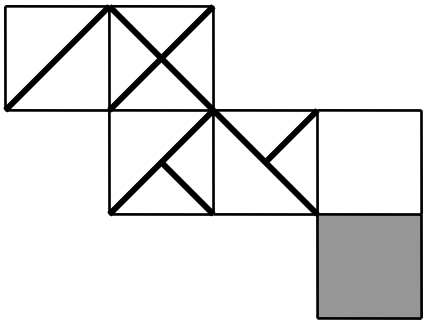
18. The digits 1, 2, 3, 4, 5, 6, 7, 8 and 9 are arranged in a circle, where every chain of three consecutive digits in the clockwise direction forms a three-digit number. What is the maximum value of the sum of all nine such numbers?

（A）4551　（B）4662　（C）4773　（D）4884　（E）4995

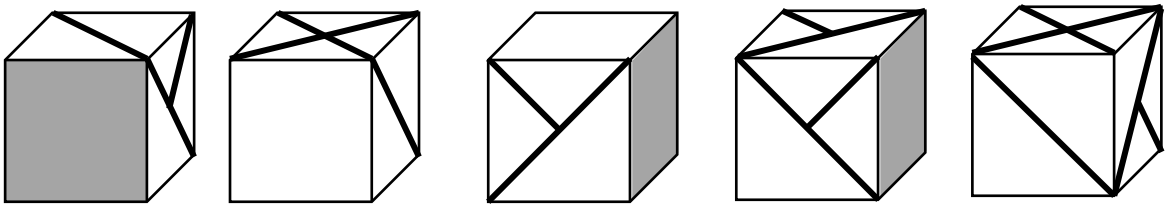
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19. For her birthday, Alice brought a box of candies, which she wanted to share with her classmates. If she gave each child 2 candies, there would be 6 candies left in the box. If she gave each child 3 candies, then exactly 5 children would be left without candies. How many candies were there in the box?
（A）22　　（B）28　　（C）33　　（D）42　　（E）48

20. The diagram below shows the net of an unfolded cube.

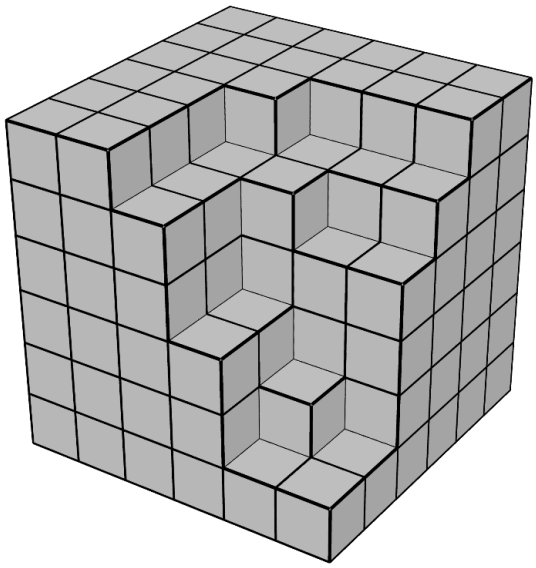


Which of the five cubes below it is not a made-up version of the flattened one?
(A)　　(B)　　(C)　　(D)　　(E)

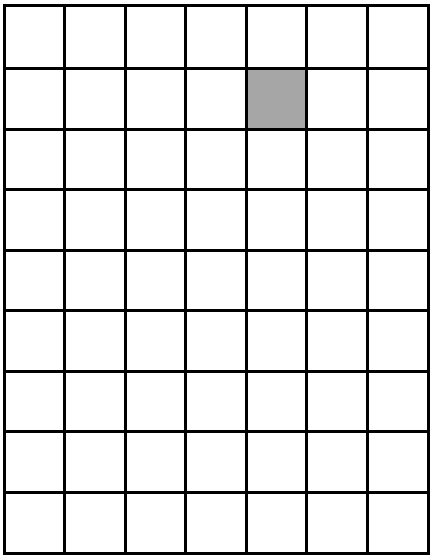


Questions 21-25, 6 marks each

21. The diagram below shows a partially built $6\times6\times6$ cube. How many unit cubes are missing?



22. A 9×7 grid of squares with one shaded square is shown in the diagram below.

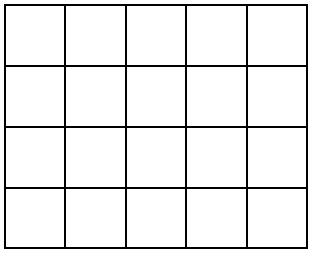


How many different rectangles bounded by the gridlines can we construct so that all rectangles contain the shaded square? (Note: a square is also a rectangle.)

23. When one of ten consecutive positive integers is removed, the sum of the remaining nine is 2022. Which number is removed?

24. The ages of Tom and his two grandchildren add up to 80 years. The ages of the two grandchildren are the digits of the grandpa's age. How old is grandpa Tom?

25. Each number in a 4×5 table is 0, 1 or 2. The sum of the numbers in each row and in each column is a multiple of 3. What is the maximum number of 1's in this table?



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